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SMA 2101: Calculus I

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke.



LESSON 4

De L' Hopital's rule Examples

$$(1) \quad \lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}, \quad f'(x) = 3x^2, \quad g'(x) = 1$$

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{3x^2}{1} = 3$$

$$(2) \quad \lim_{x \rightarrow 1} \frac{1 - \sqrt{x}}{1 - x}, \quad \lim_{x \rightarrow 1} \frac{-\frac{1}{2\sqrt{x}}}{-1} = \frac{1}{2}$$

$$(3) \quad \lim_{x \rightarrow 0} \left(\frac{x - \sin x}{x^3} \right) = \lim_{x \rightarrow 0} \frac{1 - \cos x}{3x^2} = \lim_{x \rightarrow 0} \frac{\sin x}{6x} = \lim_{x \rightarrow 0} \frac{\cos x}{6} =$$

$\frac{1}{6}$

$$(4) \quad \lim_{x \rightarrow 0} \frac{x}{\sin 2x} = \lim_{x \rightarrow 0} \frac{1}{2 \cos 2x} = \frac{1}{2}$$



$$(5) \quad \lim_{x \rightarrow 0} \frac{\sin 7x}{5x} = \lim_{x \rightarrow 0} \frac{7 \cos 7x}{5} = \frac{7}{5}$$

More Revision Questions

Evaluate

i. $\lim_{x \rightarrow 0} \frac{\sin 7x}{5x}$

ii. $\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta}$

iii. $\lim_{\theta \rightarrow 0} \frac{\sin^2 \theta}{\theta^2}$

Chain rule Differentiation

If $y = f(x)$ and $x = g(t)$, then $\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx}$ i.e

Example . $y = u^3$ and $u = 4x^2 - 2x + 5$, find $\frac{dy}{dx}$

Solution: $\frac{dy}{du} = 3u^2$, $\frac{du}{dx} = 8x - 2$, $\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = 3u^2 (8x - 2)$



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$$= 3(4x^2 - 2x + 5)^2(8x - 2)$$

□

Example . $y = x^2 - 4x$, $x = \sqrt{2t^2 + 1}$, find $\frac{dy}{dt}$ at $t = \sqrt{2}$

Solution: Let $w = 2t^2 + 1$, $x = \sqrt{w}$ then $\frac{dw}{dt} = 4t$

$$\frac{dy}{dx} = 2(x - 2), \quad \frac{dx}{dw} = \frac{1}{2}w^{-1/2}$$

$$\begin{aligned} \frac{dy}{dt} &= \frac{dy}{dx} \frac{dx}{dw} \frac{dw}{dt} = 2(x - 2) \left(\frac{1}{2}w^{-1/2}\right) 4t = (x - 2) \frac{4t}{\sqrt{w}} \\ &= (\sqrt{2t^2 + 1} - 2) \frac{4t}{\sqrt{2t^2 + 1}} = (\sqrt{5} - 2) \left(\frac{4\sqrt{2}}{\sqrt{5}}\right) \end{aligned}$$

□

More Revision Questions

Evaluate/use chain rule $\frac{dy}{dx}$

$$y = u^3 + 4; \quad u = x^2 + 2$$





$$y = \sqrt{1+u}$$

$$u = \sqrt{x}$$

$$y = \sqrt{u}$$

$$u = v(3-2v),$$

$$v = x^2$$

$$y = u^2 + 4u$$

$$u = t^{1/2} - t^{1/4}$$

Examples on Product Rule

$$\frac{d}{dx} [f(x) g(x)] = f(x) \frac{d}{dx} g(x) + g(x) \frac{d}{dx} f(x) \text{ or}$$

$$\frac{d}{dx} [uv] = u \frac{dv}{dx} + v \frac{du}{dx}$$

Example Differentiate the expression $y = (x^2 - 3)(x + 1)^2$

Solution: Let $u = x^2 - 3$ and $v = (x + 1)^2$

$$\frac{du}{dx} = 2x \quad \text{and} \quad \frac{dv}{dx} = 2(x + 1)$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx} = (x^2 - 3) 2(x + 1) + (x + 1)^2 2x$$

$$= 2(x + 1)(x^2 - 3 + x^2 + x) = 2(x + 1)(2x^2 + x - 3) =$$

$$2(x + 1)(2x + 3)x - 1$$

□





Example . Differentiate $y = (x^2 + 1)^3 (x^3 + 1)^2$

Solution: Let $u = (x^2 + 1)^3$ and $v = (x^3 + 1)^2$

$$\begin{aligned}\frac{du}{dx} &= 3(x^2 + 1)^2 2x \quad \text{and} \quad \frac{dv}{dx} = 2(x^3 + 1) 3x^2 \\ \frac{dy}{dx} &= u \frac{dv}{dx} + v \frac{du}{dx} = (x^2 + 1)^3 2(x^3 + 1) 3x^2 + (x^3 + 1)^2 3(x^2 + 1)^2 2x \\ &= 6x^2 (x^2 + 1)^3 (x^3 + 1) + 6x (x^3 + 1)^2 (x^2 + 1)^2 \\ &= 6x (x^3 + 1) (x^2 + 1)^2 [x(x^2 + 1) + (x^3 + 1)] \\ &= 6x (x^3 + 1) (x^2 + 1)^2 (2x^3 + x + 1)\end{aligned}$$

□

Example . Differentiate $(x - 3)^2 (x + 2)^{-3}$

Solution: Let $u = (x - 3)^2$ and $v = (x + 2)^{-3}$

$$\begin{aligned}\frac{du}{dx} &= 2(x - 3) \quad \text{and} \quad \frac{dv}{dx} = -3(x + 2)^{-4} \\ \frac{dy}{dx} &= u \frac{dv}{dx} + v \frac{du}{dx} = (x - 3)^2 (-3)(x + 2)^{-4} + (x + 2)^{-3} 2(x - 3) \\ &= \frac{-3(x-3)^2}{(x+2)^4} + \frac{2(x-3)}{(x+2)^3} = \frac{-3(x-3)^2 + 2(x-3)(x+2)}{(x+2)^4}\end{aligned}$$





$$= \frac{(x-3)[-3(x-3)+2(x-3)(x+2)]}{(x+2)^4}$$

□

Examples on Quotient rule

If $y = \frac{f(x)}{g(x)}$ then $\frac{dy}{dx} = \frac{1}{g(x)g'(x)} (g(x)f'(x) - f(x)g'(x))$ (Quotient rule) or

if $u = f(x)$ and $v = g(x)$ then $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$ (Quotient rule)

Example  Differentiate $y = \frac{x+1}{x^2-2}$

Solution: Let $u = x+1$ and $v = x^2-2$ then $\frac{du}{dx} = 1$ and $\frac{dv}{dx} = 2x$

$$\frac{dy}{dx} = \frac{(x^2-2)(1) - (x+1)2x}{(x^2-2)^2} = \frac{-x^2-2x-2}{(x^2-2)^2}$$

□

Example  Differentiate $y = \frac{2x-1}{x+2}$



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Solution: Let $u = 2x - 1$ and $v = x + 2$ then $\frac{du}{dx} = 2$ and $\frac{dv}{dx} = 1$

$$\frac{dy}{dx} = \frac{(x+2)(2) - (2x-1)(1)}{(x+2)^2} = \frac{2x+4-2x+1}{(x+2)^2} = \frac{5}{(x+2)^2}$$

□

More Revision Questions

1. iFind $\frac{dy}{dx}$ if (i) $x = y\sqrt{1-y^2}$ (ii) $x = y^2 - 4y$

EXERCISE 1. Find $\frac{dy}{dx}$ in each case

i. $y = \sqrt{x^2 + 6x + 3}$

ii. $y = \frac{3-2x}{3+2x}$

iii. $y = \frac{x^2}{\sqrt{4-x^2}}$

Example Find $\frac{dy}{dx}$ if $y = \frac{u^2-1}{u^2+1}$ and $u = \sqrt[3]{x^2+2}$, find $\frac{dy}{dx}$

Solution: $\frac{dy}{dx} = \frac{4u}{(u^2+1)}$ and $\frac{du}{dx} = \frac{2x}{3(x^2+2)^{2/3}} = \frac{2x}{3u^2}$

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = \frac{4u}{(u^2+1)} \frac{2x}{3u^2} = \frac{8x}{3u(u^2+1)}$$





EXERCISE 2. Differentiate

- a. If $\theta = \frac{3r+2}{2r+3}$
- b. $y = 2x^2\sqrt{2-x}$
- c. $y = \frac{\sqrt{x-1}}{\sqrt{x+1}}$
- d. $y = (x-1)\sqrt{x^2-2x+2}$
- e. $y = x\sqrt{3-2x^2}$



Solutions to Exercises

Exercise 1. i. $\frac{dy}{dx} = \frac{x+3}{\sqrt{x^2+6x+3}}$

ii. $\frac{dy}{dx} = \frac{-12}{(3+2x)^2}$

iii. $\frac{dy}{dx} = \frac{8x-x^3}{(4-x^2)^{3/2}}$

Exercise 1



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Exercise 2. a. $\frac{d\theta}{dr} = \frac{5}{(2r+3)^2}$ b. $y' = \frac{x(8-5x)}{\sqrt{2-x}}$

c. $y' = \frac{1}{(x+1)\sqrt{x^2-1}}$

d. $\frac{dy}{dx} = \frac{2x^2-4x+3}{x^2-2x+2}$

e. $y' = \frac{3-4x^2}{\sqrt{3-2x^2}}$

Exercise 2